Differential Equations & Transform Techniques.

(I) Logistics

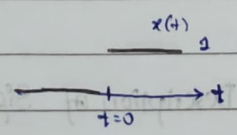
Exam Slots : 03 Feb
24 Feb 6:30 - 8:30 pm
24 Mar
09 Apr
30 Apr

(II) Motivation

Functions, signals or waveforms appear in a wide variety of engineering applications and carry information about the behaviour of some phenomenon of interest.

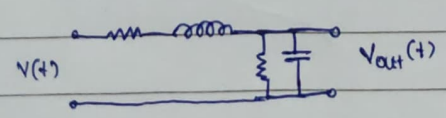
Typically, a signal is described as a function of time parameter (t).
signal x : value of $x(t)$ at time t.

$$x(t) = \begin{cases} 0 , & t < 0 \\ 1 , & t \geq 0 \end{cases}$$



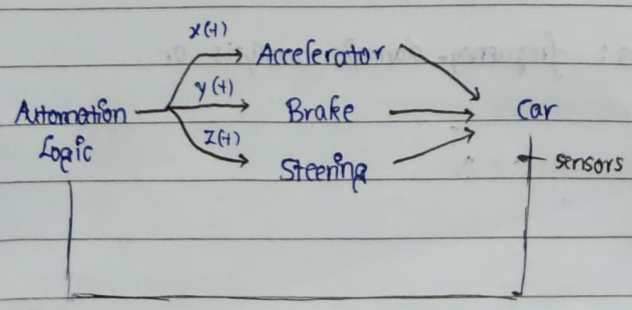
Examples :-

(1)



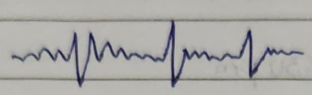
} circuit analysis

(2)



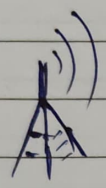
③ Signal Processing

Reading of an electrocardiogram



→ carries info on coronary health

④ Communications



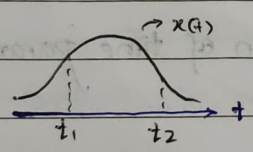
electromagnetic wave → modelled as a single / waveform

↓

carries information from radio / TV station / channels.

⑤ Probability Theory

$x(t)$ = probability density function



$\int_{t_1}^{t_2} x(t)dt$ = probability that the random variable takes values from interval t_1 to t_2.

Transform-domain Description of Signal:

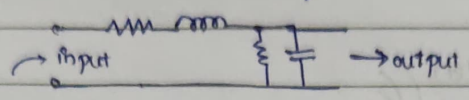
Main idea: We express, either exactly or approximately, the signal of interest to us $x(t)$ as a linear continuition of sinusoidal waveforms.

Also known as: frequency-domain analysis or

Examples:

① Circuit Analysis.

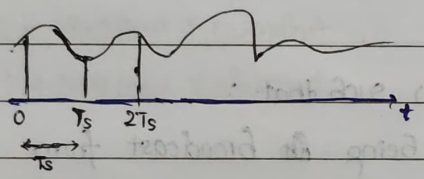
a "linear" time invariant system



$\sum_i \alpha_i \sin(2\pi f_i t)$ → $\sum_i \alpha_i A_i \sin(2\pi f_i t + \phi_i)$

A_i, ϕ_i are constants that depend on the input freq and the circuit.

② Signal processing



T_s = sampling interval

Can we retrieve the value of $x(t)$ at all t, given the values of the samples $x(0)$, $x(T_s)$, $x(2T_s)$, ...

⇓

Answered by Sampling theorem.

③ Probability theorem

$$\frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}} \rightarrow \text{Guassian distribution as } n \to \infty$$

central limit theorem

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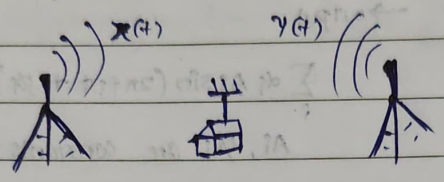
The reasoning / proof uses transform-domain analysis

↓ moment generating function (MGF)

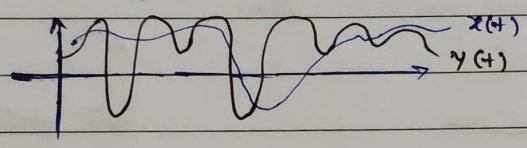
Communications

In practice, multiple wireless transmitters broadcast information simultaneously.



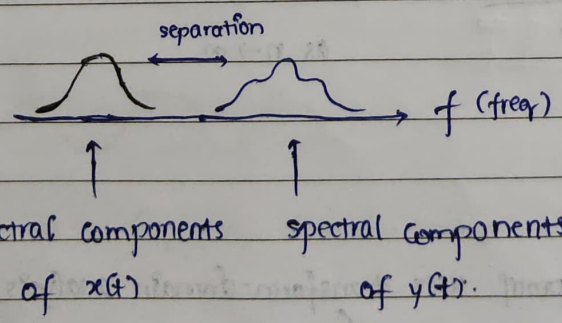
The receiver observes the sum of $x(t)$ and $y(t)$

$$r(t) = x(t) + y(t)$$



Can we design $x(t)$ and $y(t)$ such that :

1. they carry the information being broadcast from the two base stations, respectively ; and
2. $x(t)$ and $y(t)$ can be separated from each other at the receiver ?

Answer: Yes,

A popular solution to this problem is called Frequency - Division Multiple Access (FDMA) whereas $x(t)$ and $y(t)$ are separated in the frequency domain.